Engineering Chemistry (AHT-002) - Unit 4

Unit 4 - Materials: Polymers, Fuels, Lubricants, Glass - notes in easy English, matched to the UTU syllabus.

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What this unit covers (syllabus): materials - polymers, fuels, lubricants and glass/cement basics.

Key ideas: Polymers: addition (chain-growth - polyethylene, PVC) vs condensation (step-growth - nylon, bakelite, with a small molecule eliminated). Thermoplastics soften on heating (recyclable); thermosets set permanently. **Fuels:** calorific value (higher and lower), characteristics of a good fuel, knocking and **octane number** for petrol, **cetane number** for diesel; gaseous fuels (CNG, LPG, biogas) compared on calorific value. **Lubricants:** reduce friction and wear; classified solid/semi-solid/liquid with the key properties (viscosity, viscosity index, flash point) - full details in the notes written under this unit. **Glass:** a supercooled liquid of silicates; types (soda-lime, borosilicate) and annealing.

Exam pointers: "addition vs condensation polymerisation with two examples each" and "thermoplastic vs thermoset" tables are free marks. Octane vs cetane numbers and "properties of a good lubricant" are the short-answer regulars.

Extra note - Lubricants - quick exam notes:

Lubricant: a substance between two moving surfaces that reduces friction and wear.

Functions: reduces friction and wear, carries away heat, seals out dust, prevents corrosion, and cushions shock between parts.

Classification: (1) **Solid** - graphite, molybdenum disulphide (used where oils would catch fire or leak); (2) **Semi-solid** - greases (oil + soap thickener; for slow, heavy loads); (3) **Liquid** - mineral oils (petroleum), vegetable and animal oils, blended oils (mineral oil + additives for viscosity index, oxidation stability).

Key properties: viscosity and viscosity index (how much viscosity changes with temperature), flash and fire point, cloud and pour point (low-temperature flow), oiliness (sticking to the surface), acid value, saponification value.

Exam tip: definitions + one example each of solid/semi-solid/liquid lubricant, and the meaning of viscosity index - this pattern repeats in papers.

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